

Surcomplex Spectral Theory

Singular Values, Infinitesimal Rank,
and Multiscale Stability

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Abstract

Finite matrices over the surcomplex numbers admit much of the familiar spectral theory, but its justification is algebraic rather than compactness based, and its interpretation changes when singular values occupy different infinitesimal scales. We develop Hermitian and normal diagonalization, positive square roots, singular value and polar decompositions, variational principles, least squares, low-rank approximation, conditioning, and spectral-subspace perturbation over a real closed field and its complexification.

The central synthesis identifies the valuations of singular values with successive differences of the minimum valuations of minors. A positive Cauchy–Binet identity then recovers both these scales and the leading singular coefficients from the characteristic coefficients of the Gram matrix. This gives an intrinsic rank filtration, an exact root-free scale algorithm, and precision certificates valid for arbitrary-rank Hahn value groups. Worked examples separate exact rank from standard-part rank, small perturbations from gap-relative perturbations, and algebraic inverses from Hahn-summable inverse series. Nonnormal pseudospectra, regularization, and a finite tensor application complete the picture.

Status and scope. A self-contained research exposition relative to standard surreal normal-form and Hahn-field foundations. Classical results are identified as such; no priority claim or solution of a named open problem is made. All matrix dimensions are ordinary finite integers. General proofs are not proof-assistant verified. The accompanying exact symbolic checks exercise finite examples, not arbitrary Hahn supports or the general theorems.

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1 The documentation gap and the mathematical agenda

1.1 What was inspected

This article was chosen after inspecting the documentation catalogue and relevant inventories in the repository [VladimirReshetnikov/Surreal](#), at the fixed revision

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The catalogue lists fifteen packages in surreal mathematics, surcomplex analysis and geometry, and foundations and computation [1]. The detailed polynomial-algebra inventory and its opening source sections were especially relevant [2]; the foundations inventory was also inspected [3]. This is a targeted coverage audit, not a claim to have checked every line of all archived source manuscripts.

Matrix theory is not wholly absent. The polynomial report uses an algebraic Schur inequality and a differentiating compression in its treatment of Schoenberg’s inequality, and it uses signatures of Hermite forms to count real roots. Finite residue pairings and multiplication matrices also appear in the surrounding algebra. These are substantial ingredients, but they are organized around polynomial roots and residues, not around matrices as the objects of study. The inspected inventories do not offer a systematic singular-value, conditioning, or scale-resolved rank theory.

Existing strand	What it supplies	What this chapter adds
Polynomial algebra	Root geometry, Schur-type arguments, signatures, Newton profiles	Unitary geometry, SVD, positive Gram matrices, determinantal singular scales
Analytic geometry	Finite algebras and deformation questions	Gap-sensitive spectral projections and Schur-complement effective matrices
Foundations	Set-sized Hahn workspaces; limits on transfer	Explicit finite algebraic proofs without compactness or a class-sized choice principle
Computer algebra	Representation and precision distinctions	Root-free scale elimination and certificates for resolvable rank

Table 1: The gap is an organizing theory, not the absence of every matrix lemma.

1.2 Why this is more than replacing scalars by surreals

Over the ordinary reals, a positive singular value is often tacitly regarded as a non-negligible number. Over the surreals, a matrix can have singular values at scales 1 , ω^{-2} , and $\omega^{-\omega}$. It is then simultaneously invertible, singular after taking standard parts, and indistinguishable from a lower-rank matrix at many prescribed precisions. An infinitesimal perturbation can rotate an eigenspace through an ordinary angle when its spectral gap is infinitesimal too. Conversely, an infinite entry need not imply an unstable direction if every singular value has the same scale.

The issue is therefore not just whether a decomposition exists. We need to know what its invariants measure, which conclusions survive reduction to an ordinary complex matrix, and which operations can be certified without computing an entire Hahn expansion. The key bridge is

$$\min_{|I|=|J|=k} v(\det A_{I,J}) = \sum_{j=1}^k v(\sigma_j(A)), \quad 1 \leq k \leq \text{rank } A. \quad (1.1)$$

Here the singular values are ordered by their actual positive magnitudes, whereas the valuations

are ordered in the opposite sense. This identity turns analytic-looking size information into finite determinantal algebra.

1.3 Attribution and reading route

The spectral theorem over real closed fields is classical, not a novel surreal phenomenon; a recent explicit use is Wang [6]. The variational and perturbation principles have their classical matrix-analysis counterparts, for example in Tao’s lecture notes [7], the original low-rank approximation paper of Eckart and Young [8], and the Davis–Kahan literature [10, 11]. The pseudoinverse is the classical Moore–Penrose construction [9].

The contribution here is an integrated account with proofs in the exact field setting, the determinantal and Gram-coefficient description of surreal scales, and explicit boundaries for reduction, computation, and summability. We do not claim that these consequences of existing algebra first occur here. Readers already comfortable with real-closed-field linear algebra can start at [Section 7](#); readers interested primarily in computation can proceed from there to [Sections 10](#) and [11](#).

2 Fields, workspaces, and conventions

2.1 The algebraic setting

Until [Section 7](#), let F be a set-sized real closed ordered field and put $K = F(i)$, where $i^2 = -1$. We use the standard equivalent characterization that F is ordered and $F(i)$ is algebraically closed. Conjugation is $\overline{x + iy} = x - iy$, and

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in F_{\geq 0}.$$

Every positive element of F has a unique positive square root. No Archimedean property or completeness assumption is imposed.

For a matrix A , write $A^* = \bar{A}^T$. A square matrix is *Hermitian* when $A^* = A$, *normal* when $A^*A = AA^*$, and *unitary* when $A^*A = AA^* = I$. A matrix with real entries is treated inside F , with transpose in place of conjugate transpose. The integers m, n, k used as dimensions and ranks are always ordinary finite integers. Ambient row and column sizes are positive; zero-dimensional invariant subspaces and empty products are allowed. Unless explicitly stated otherwise, vector spaces and subspaces are over K .

We use ∞ only as an added valuation symbol, with $v(0) = \infty$. It is not a surreal number. Expressions such as $\infty - \infty$ will never be used.

2.2 Hahn workspaces inside the surcomplex numbers

For the scale-sensitive results, choose a divisible set-sized additive subgroup $\Gamma \subseteq \mathbf{No}$, with its inherited order, and set

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)), \quad t^\gamma := \omega^{-\gamma}. \quad (2.1)$$

The last expression uses Conway’s monomial map. It is not a convention about the ordinary real exponential. An element of K_Γ is a formal sum

$$a = \sum_{\gamma \in S} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{C},$$

where $S \subseteq \Gamma$ is a well-ordered set in increasing exponent order. Zero coefficients may be omitted. The field operations are Hahn addition and convolution. On F_Γ , the sign of a nonzero series is the sign of its coefficient at the least exponent.

We use two foundational facts. Conway normal forms identify these series with surcomplex numbers, and the Hahn field with divisible value group and algebraically closed coefficient field of characteristic zero is algebraically closed. Gonshor's Chapter 5 gives the normal-form and support foundations [4]; Poonen's Corollary 4 gives the relevant algebraic-closedness statement for Mal'cev–Neumann fields [5]. Thus $K_\Gamma = F_\Gamma(i)$ is algebraically closed and F_Γ is real closed. These standard foundational theorems are inputs, not reproved here.

Proposition 2.1 (Finite-data localization). *Every finite collection of surcomplex matrices has all its entries in one workspace (2.1). All the decompositions established below can be performed in that workspace.*

Proof. Take the union of the exponent supports in the real and imaginary normal forms of every entry. There are finitely many entries and every support is a set, so their union is a set. Its rational linear span is a divisible set-sized subgroup of $\mathbf{No}$. Reversing exponents to follow the convention $t^\gamma = \omega^{-\gamma}$ puts all entries in its Hahn field. Algebraic closedness supplies all roots needed in the spectral proofs; positive square roots and finite field operations remain in the same workspace. Every basis extension used below has finite length. $\square$

The same argument works for a set of scalar inputs, but we only need finite matrices here. No quantification over a class-sized set of all bases is required. A displayed theorem about matrices over $\mathbf{No}[i]$ means that the finite data are localized and the theorem is applied there. Its explicit algebraic witnesses are still witnesses in any larger workspace.

2.3 Order, valuation, and topology are separate structures

In a Hahn workspace the ordered modulus and the valuation are not the same thing. For $\rho > 0$, both t^ρ and $2t^\rho$ have valuation ρ , although the latter is outside the ordered disk $|z| \leq t^\rho$. We will use ordered inequalities for optimization and conditioning, and valuation inequalities for relative scale and support precision.

The principal proofs in this article use no infinite limiting process. A unit sphere is not assumed compact, a supremum is not assumed to exist, and no spectral integral is invoked. In Section 15, where an infinite matrix series is used, its Hahn support is checked explicitly. Internal valuation convergence in a chosen Hahn field and the fine order/modulus topology inherited from the whole surreal class must not be conflated.

3 Hermitian geometry without compactness

For $x, y \in K^n$, define

$$\langle x, y \rangle = x^* y = \sum_{j=1}^n \bar{x}_j y_j, \quad \|x\| = \sqrt{x^* x}.$$

This convention is conjugate-linear in the first variable. Since F is ordered, a sum of squares is zero only if every term is zero. Consequently $x^* x > 0$ for $x \neq 0$.

Lemma 3.1 (Cauchy–Schwarz and orthogonal decomposition). *For $x, y \in K^n$,*

$$|x^* y| \leq \|x\| \|y\|.$$

If $x \neq 0$, the vector y decomposes uniquely as

$$y = x \frac{x^*y}{x^*x} + y_\perp, \quad x^*y_\perp = 0.$$

Every finite-dimensional subspace admits an orthonormal basis and has an orthogonal complement.

Proof. The asserted decomposition follows by substitution. Its squared norm is

$$y^*y = \frac{|x^*y|^2}{x^*x} + y_\perp^*y_\perp \geq \frac{|x^*y|^2}{x^*x},$$

which proves Cauchy–Schwarz; the case $x = 0$ is immediate. Gram–Schmidt now works on any finite linearly independent list: subtract the finitely many orthogonal components already constructed and divide the remaining nonzero vector by its positive norm. Extend a basis of the subspace to a basis of K^n and continue this process. The remaining orthonormal vectors span its orthogonal complement. $\square$

This proof still works when a norm is infinitesimal. Dividing by that norm may produce infinite coordinates at an intermediate step, but it is a legal field operation; the final unit vector has all coordinate moduli at most one. It is invalid to delete a nonzero vector merely because its standard part vanishes.

The triangle inequality follows from Cauchy–Schwarz:

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2.$$

Finite orthogonal projections therefore exist by an explicit basis formula, not by an infinite-dimensional projection theorem. If $u_1, \dots, u_k$ is an orthonormal basis of W , then

$$P_W = \sum_{j=1}^k u_j u_j^*, \quad P_W^2 = P_W = P_W^*, \quad K^n = W \oplus W^\perp. \quad (3.1)$$

The projector is independent of the chosen orthonormal basis because it acts as the identity on W and as zero on $W^\perp$.

4 Spectral, positive, and singular-value decompositions

4.1 The finite spectral theorem

Theorem 4.1 (Hermitian spectral theorem). *For every Hermitian $H \in K^{n \times n}$, there are a unitary U and $\lambda_1, \dots, \lambda_n \in F$ such that*

$$H = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*.$$

For real symmetric H , U can be chosen orthogonal with entries in F .

Proof. The characteristic polynomial splits in K , so there is an eigenpair $Hx = \lambda x$ with $x \neq 0$. The scalar x^*Hx is fixed by conjugation, hence lies in F , and

$$\lambda = \frac{x^*Hx}{x^*x} \in F.$$

Normalize x . If $y \perp x$, then $x^*Hy = (Hx)^*y = \lambda x^*y = 0$, so $x^\perp$ is invariant under H . The restriction to that complement is Hermitian. Induction on n completes the construction. For a real symmetric matrix, once $\lambda \in F$ is known, singularity of $H - \lambda I$ over F supplies a nonzero kernel vector in F^n . The same induction can therefore be carried out over F . $\square$

Proposition 4.2 (Schur form and normal matrices). *Every $A \in K^{n \times n}$ is unitarily similar to an upper triangular matrix. It is unitarily diagonalizable if and only if it is normal.*

Proof. Choose and normalize an eigenvector and extend it to an orthonormal basis. The first column below the diagonal then vanishes. Apply induction to the remaining square block to obtain upper triangular Schur form T . If T is normal, comparison of the $(1, 1)$ entries of TT^* and T^*T gives

$$|t_{11}|^2 + \sum_{j>1} |t_{1j}|^2 = |t_{11}|^2.$$

Positivity forces the rest of the first row to vanish. Induction on the remaining normal block makes T diagonal. Conversely a diagonal matrix is normal, and unitary conjugation preserves normality. $\square$

For distinct eigenvalues λ of a Hermitian matrix, its spectral projectors are finite polynomials:

$$P_\lambda = \prod_{\substack{\mu \in \text{spec } H \\ \mu \neq \lambda}} \frac{H - \mu I}{\lambda - \mu}, \quad H = \sum_{\lambda \in \text{spec } H} \lambda P_\lambda. \quad (4.1)$$

Evaluation on the diagonal form proves the formula. The eigenspace projector is canonical; a basis within a repeated eigenspace is not.

4.2 Positivity, square roots, and congruence

A Hermitian matrix is positive semidefinite, written $H \succeq 0$, when $x^* H x \geq 0$ for every x , and positive definite, written $H \succ 0$, when this is strict for $x \neq 0$. These are equivalent respectively to all eigenvalues being nonnegative or positive. In particular, positive definite does not mean bounded below by a positive *ordinary real* multiple of I .

Proposition 4.3 (Positive square root and inertia). *A positive semidefinite matrix has a unique positive semidefinite square root. The numbers of positive, negative, and zero eigenvalues of a Hermitian matrix are unchanged by an invertible congruence $H \mapsto S^* H S$.*

Proof. Diagonalize H and replace each eigenvalue by its nonnegative square root. For uniqueness, a positive semidefinite R with $R^2 = H$ commutes with H , so it preserves each eigenspace of H . On the eigenspace with eigenvalue λ , diagonalization of R gives only nonnegative roots of $r^2 = \lambda$, hence $R = \sqrt{\lambda} I$ there.

If p is the number of positive eigenvalues of H , its positive eigenspace is a p -dimensional subspace on which the quadratic form is positive definite. Any subspace of dimension greater than p meets the span of the nonpositive eigenvectors nontrivially, so cannot have that property. Thus p is an intrinsic maximum dimension, preserved by invertible congruence. Apply the same argument to $-H$ for the negative count; subtract both counts from n for the nullity. $\square$

The usual Cholesky factorization also survives. If $H = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \succ 0$, then $a > 0$ and $C - bb^*/a \succ 0$: evaluate the form at $(-a^{-1}b^*y, y)$. Inductively factor that Schur complement as LL^* . Then

$$H = \begin{pmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & L \end{pmatrix} \begin{pmatrix} \sqrt{a} & b^*/\sqrt{a} \\ 0 & L^* \end{pmatrix}.$$

This is a finite algorithm, not a limiting process.

4.3 Singular values and the polar decomposition

Theorem 4.4 (Singular value decomposition). *Let $A \in K^{m \times n}$ and $p = \min(m, n)$. There are unitaries U, V and a rectangular diagonal matrix Σ such that*

$$A = U\Sigma V^*, \quad \sigma_1 \geq \cdots \geq \sigma_r > 0, \quad \sigma_{r+1} = \cdots = \sigma_p = 0,$$

where $r = \text{rank } A$ and the diagonal entries of Σ are the σ_j . The positive σ_j are uniquely determined with multiplicity.

Proof. The matrix A^*A is positive semidefinite. Choose an orthonormal eigenbasis $v_1, \dots, v_n$ and write its positive eigenvalues as $\sigma_1^2, \dots, \sigma_r^2$. For $j \leq r$, put $u_j = Av_j/\sigma_j$. Then

$$u_j^* u_k = (\sigma_j \sigma_k)^{-1} v_j^* A^* A v_k = \delta_{jk}.$$

When the eigenvalue is zero, $\|Av_j\|^2 = 0$ implies $Av_j = 0$. Extend the u_j to an orthonormal basis of K^m . These bases give the claimed factorization, with precisely r nonzero diagonal entries. The characteristic polynomial of A^*A determines the squared singular values, and the positive square root determines the singular values themselves. $\square$

When convenient we extend the list by zeros up to length n , so that σ_j^2 lists all eigenvalues of A^*A . In particular, formulas using input subspaces of dimension k remain meaningful for $k > p$.

Set $|A| = (A^*A)^{1/2}$. The SVD gives the polar factorization

$$A = W|A|, \quad W = \sum_{j=1}^r u_j v_j^*. \quad (4.2)$$

Here W is the unique map that is an isometry from $(\ker A)^\perp$ to $\text{im } A$, is zero on $\ker A$, and satisfies (4.2). On that complement $|A|$ is invertible, so the uniqueness follows directly. For a square invertible matrix, W is unitary. For a singular square matrix, unitary extensions of W exist but are generally not unique. Confusing the canonical partial isometry with one of those extensions loses information.

4.4 A generalized Hermitian eigenproblem

If $G \succ 0$ and $H = H^*$, the equation $Hx = \lambda Gx$ reduces to the ordinary Hermitian problem for

$$C = G^{-1/2} H G^{-1/2}, \quad y = G^{1/2} x.$$

Consequently every generalized eigenvalue is in F , and there is a basis x_j with $x_j^* G x_k = \delta_{jk}$. This is useful when the metric itself contains infinitesimal weights. It does not require a uniform ordinary-real lower bound on G .

5 Norms and variational principles with attained extrema

5.1 Operator and Frobenius norms

Define

$$\|A\|_{\text{op}} = \sigma_1(A), \quad \|A\|_{\text{F}} = \sqrt{\text{tr}(A^*A)} = \sqrt{\sum_{i,j} |a_{ij}|^2},$$

with operator norm zero for the zero matrix. The SVD proves, rather than assumes, that

$$\|A\|_{\text{op}} = \max_{\|x\|=1} \|Ax\|. \quad (5.1)$$

Indeed, writing $x = Vy$ gives $\|Ax\|^2 = \sum_j \sigma_j^2 |y_j|^2 \leq \sigma_1^2$, with equality at a leading right singular vector. Thus the symbol “max” is legitimate even though the sphere was never declared compact.

Both norms are invariant under compatible unitary multiplication. They satisfy triangle inequalities, and

$$\max_{i,j} |a_{ij}| \leq \|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq \sqrt{mn} \max_{i,j} |a_{ij}|, \quad (5.2)$$

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}, \quad \|AB\|_{\text{F}} \leq \|A\|_{\text{op}} \|B\|_{\text{F}}. \quad (5.3)$$

For the first inequality in (5.2), apply A to a coordinate unit vector and then bound one coordinate by the vector norm. The middle inequality follows from $\sigma_1^2 \leq \sum_j \sigma_j^2$, and the last one from the finite sum defining the Frobenius norm. The operator product bound follows from (5.1); the Frobenius bound follows by applying it to every column of B and summing squares. The companion inequality $\|AB\|_{\text{F}} \leq \|A\|_{\text{F}} \|B\|_{\text{op}}$ follows by taking adjoints.

5.2 Courant–Fischer and Ky Fan principles

Theorem 5.1 (Variational principles). *Let $H = H^*$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. For $1 \leq k \leq n$,*

$$\lambda_k = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} x^* H x, \quad (5.4)$$

$$= \min_{\dim W=n-k+1} \max_{\substack{x \in W \\ \|x\|=1}} x^* H x. \quad (5.5)$$

Also, for $0 \leq k \leq n$,

$$\sum_{j=1}^k \lambda_j = \max_{X^* X = I_k} \text{tr}(X^* H X). \quad (5.6)$$

All inner and outer extrema in these statements are attained.

Proof. For a fixed subspace W with orthonormal basis matrix X , the compression $X^* H X$ is Hermitian. Its largest and smallest eigenvalues are the extrema of the restricted Rayleigh quotient, by diagonalization. This proves inner attainment without compactness.

Every k -dimensional W meets the span of the eigenvectors for $\lambda_k, \dots, \lambda_n$ in a nonzero vector. After normalization that vector has Rayleigh quotient at most λ_k . The span of the first k eigenvectors attains the opposite bound, proving (5.4). The complementary dimension argument proves (5.5).

For (5.6), express the columns of X in an orthonormal eigenbasis of H . If a_j is the sum of their squared moduli in row j , then $0 \leq a_j \leq 1$ and $\sum_j a_j = k$. Hence

$$\text{tr}(X^* H X) = \sum_j a_j \lambda_j \leq \sum_{j=1}^k \lambda_j.$$

To see the inequality even for signed λ_j , subtract λ_k from every λ_j . The terms before k have nonnegative coefficients and can only decrease on replacing 1 by a_j ; the terms after k are nonpositive. The constant part is unchanged because the weights sum to k . The first k eigenvectors attain equality. $\square$

Applying (5.4) to $A^* A$ and taking nonnegative square roots gives the singular-value principle

$$\sigma_k(A) = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} \|Ax\|. \quad (5.7)$$

The list is padded with zeros to length n when necessary. Applying (5.6) to $-H$ similarly gives the minimum trace as the sum of the smallest k eigenvalues. These formulas are algebraic versions of familiar classical results [7]; their finite-dimensional proofs explain why they remain valid over F .

5.3 Perturbing eigenvalues and singular values

Theorem 5.2 (Ordered Lipschitz bounds). *If H, E are Hermitian of the same size, and both eigenvalue lists are ordered decreasingly, then*

$$|\lambda_j(H + E) - \lambda_j(H)| \leq \|E\|_{\text{op}}.$$

For arbitrary rectangular matrices of the same size,

$$|\sigma_j(A + E) - \sigma_j(A)| \leq \|E\|_{\text{op}}. \quad (5.8)$$

Proof. For a unit vector x , Cauchy–Schwarz gives $|x^* E x| \leq \|E\|_{\text{op}}$. Insert the two inequalities for the perturbed Rayleigh quotient into (5.4), obtaining the Hermitian result. For singular values, the triangle inequality gives $|\|(A + E)x\| - \|Ax\|| \leq \|E\|_{\text{op}}$ on the unit sphere. Take the attained minima and maxima in (5.7). $\square$

Theorem 5.3 (A Frobenius eigenvalue bound). *For Hermitian H, G with decreasing eigenvalue lists λ, μ ,*

$$\sum_{j=1}^n (\lambda_j - \mu_j)^2 \leq \|H - G\|_{\text{F}}^2.$$

Proof. Let u_i, v_j be orthonormal eigenbases and set $p_{ij} = |u_i^* v_j|^2$. The matrix p has nonnegative entries and every row and column sum is one. Put $c_i = \sum_j p_{ij} \mu_j$. For every k ,

$$\sum_{i=1}^k c_i \leq \sum_{j=1}^k \mu_j,$$

by the weight argument in Theorem 5.1: the column weights $\sum_{i \leq k} p_{ij}$ lie in $[0, 1]$ and sum to k . Equality holds for $k = n$. Summation by parts, using $\lambda_i - \lambda_{i+1} \geq 0$, now gives

$$\text{tr}(HG) = \sum_i \lambda_i c_i \leq \sum_i \lambda_i \mu_i.$$

Expand $\|H - G\|_{\text{F}}^2 = \text{tr } H^2 + \text{tr } G^2 - 2 \text{tr}(HG)$ to conclude. $\square$

This proof does not use a decomposition of a doubly stochastic matrix into permutations, a measure on a unitary group, or a real-valued limiting argument. It is useful when all the eigenvalue movements, not just the largest one, matter.

6 Least squares, pseudoinverses, and best low-rank models

6.1 The Moore–Penrose inverse

Given an SVD $A = U \Sigma V^*$, let $\Sigma^\dagger$ be the transposed rectangular diagonal matrix whose nonzero entries are σ_j^{-1} and whose remaining entries are zero. Define

$$A^\dagger = V \Sigma^\dagger U^*.$$

Direct diagonal multiplication proves the Penrose identities

$$AA^\dagger A = A, \quad A^\dagger AA^\dagger = A^\dagger, \quad (AA^\dagger)^* = AA^\dagger, \quad (A^\dagger A)^* = A^\dagger A. \quad (6.1)$$

They characterize $A^\dagger$ uniquely, as in the classical construction [9]. For completeness, if B satisfies these identities, AB is the orthogonal projector onto $\text{im } A$: it is a Hermitian idempotent with that image. Similarly BA is the orthogonal projector onto $(\ker A)^\perp$, since $\ker(BA) = \ker A$. The equation $B = BAB$ then says that B is zero on $(\text{im } A)^\perp$ and, on $\text{im } A$, is the inverse of the restriction $A : (\ker A)^\perp \rightarrow \text{im } A$. This uniquely determines it and also shows that the SVD definition is basis-independent.

Theorem 6.1 (Least squares with an attained minimum). *For $b \in K^m$, all minimizers of $\|Ax - b\|$ are*

$$x = A^\dagger b + z, \quad z \in \ker A.$$

The unique minimizer of least norm is $A^\dagger b$, and the minimum residual is $\|(I - AA^\dagger)b\|$.

Proof. Write $c = U^*b$ and $y = V^*x$. The squared residual is

$$\sum_{j=1}^r |\sigma_j y_j - c_j|^2 + \sum_{j=r+1}^m |c_j|^2.$$

It is minimized precisely by $y_j = c_j/\sigma_j$ for $j \leq r$, with remaining input coordinates free. These free coordinates describe $\ker A$. They add a nonnegative squared norm to the fixed solution, so setting them to zero uniquely minimizes the norm. The residual formula is the corresponding orthogonal projection formula. $\square$

Small singular values are not discarded in this theorem. Their reciprocals may be infinite, and a mathematically exact least-squares solution may therefore have infinite norm even when the data are finite. Regularization in [Section 16](#) changes the optimization problem rather than silently changing what “nonzero” means.

6.2 Low-rank approximation

Let

$$A_k = \sum_{j=1}^k \sigma_j u_j v_j^*, \quad 0 \leq k < p.$$

Here zero terms can be included when $k > r$.

Theorem 6.2 (Eckart–Young bounds over F). *Among matrices B of rank at most k ,*

$$\min \|A - B\|_{\text{op}} = \sigma_{k+1}(A), \quad (6.2)$$

$$\min \|A - B\|_F^2 = \sum_{j=k+1}^p \sigma_j(A)^2. \quad (6.3)$$

Both minima are attained by A_k .

Proof. The SVD computes the two errors of A_k exactly. For the lower operator bound, a matrix B of rank at most k has a kernel meeting the span of $v_1, \dots, v_{k+1}$ in a nonzero vector. Normalize that vector x . Then

$$\|(A - B)x\| = \|Ax\| \geq \sigma_{k+1}.$$

For the Frobenius bound, choose $n - k$ orthonormal vectors w_j in $\ker B$ and extend them to an orthonormal basis. Summing squared column norms in that basis gives

$$\|A - B\|_F^2 \geq \sum_{j=1}^{n-k} \|Aw_j\|^2.$$

The minimum-trace version of (5.6), applied to A^*A , bounds the right side below by the sum of its smallest $n - k$ eigenvalues. That is exactly (6.3), after padding the singular list with zeros. $\square$

The least-squares lower-rank problem originates in Eckart and Young [8]. Here its optimal error is allowed to be infinitesimal or infinite. The theorem asserts existence of an optimizer, not uniqueness: repeated singular values and the operator norm in particular can produce many optimizers. It also gives a direct operational meaning to every singular scale: σ_{k+1} is the exact error required to replace the matrix by a rank- k model.

7 Hahn valuation and singular-value scales

From this section onward, unless a result explicitly returns to arbitrary F , work in (2.1). For $a \neq 0$, put

$$v(a) = \min \operatorname{supp} a \in \Gamma, \quad \operatorname{lc}(a) = a_{v(a)}.$$

Let

$$\mathcal{O} = \{a : v(a) \geq 0\}, \quad \mathfrak{m} = \{a : v(a) > 0\}.$$

The residue map $\operatorname{st} : \mathcal{O} \rightarrow \mathbb{C}$ extracts the coefficient at exponent zero. On real finite elements it is the usual standard part. The valuation of a matrix or vector means the minimum valuation of its entries; all-zero objects have valuation ∞ . We sometimes write $v_{\operatorname{mat}}(A)$ to make this convention explicit.

7.1 Positivity prevents leading cancellation

Lemma 7.1 (Valuation of a Euclidean norm). *For any nonzero finite vector x and any nonzero matrix A ,*

$$v(\|x\|) = \min_j v(x_j), \quad v(\|A\|_{\operatorname{op}}) = v(\|A\|_F) = \min_{i,j} v(a_{ij}). \quad (7.1)$$

Proof. Let $\alpha = \min_j v(x_j)$. The coefficient of $t^{2\alpha}$ in $\sum_j |x_j|^2$ is

$$\sum_{v(x_j)=\alpha} |\operatorname{lc}(x_j)|^2 > 0.$$

Thus its valuation is 2α , and taking its positive square root gives valuation α . Apply the same argument to the entries of A for the Frobenius norm. Finally (5.2) traps the operator norm between a largest entry modulus and a fixed positive ordinary-real multiple of it. Both bounds have the same valuation, proving the remaining equality. $\square$

A finite sum of positive leading coefficients is crucial. An arbitrary sum of complex entries can cancel at its leading exponent; a squared Euclidean norm cannot.

Lemma 7.2 (Unitary matrices preserve the integral lattice). *If U is unitary, then $U, U^{-1} \in \mathcal{O}^{n \times n}$. Its reduction $\operatorname{st} U$ is an ordinary complex unitary matrix, and $U\mathcal{O}^n = \mathcal{O}^n$.*

Proof. Every entry has modulus at most one because its column has squared norm one. Hence every entry has nonnegative valuation. The inverse is U^* and has the same property. Reducing $U^*U = I$ gives $(\text{st } U)^*(\text{st } U) = I$, and the two lattice inclusions follow from integrality of U and its inverse. $\square$

For the nonzero singular values of A , define

$$\gamma_j = v(\sigma_j), \quad \gamma_1 \leq \cdots \leq \gamma_r. \quad (7.2)$$

Their order is reversed because a larger positive Hahn number has a smaller or equal leading exponent. Different singular values may have the same valuation. The tuple (7.2) records only scales, not the positive leading coefficients or the rest of each expansion.

7.2 The determinantal characterization

Define the k th determinantal valuation by

$$\Delta_0(A) = 0, \quad \Delta_k(A) = \min_{\substack{I \subseteq \{1, \dots, m\}, |I|=k \\ J \subseteq \{1, \dots, n\}, |J|=k}} v(\det A_{I,J}), \quad 1 \leq k \leq p. \quad (7.3)$$

It is ∞ exactly when $k > r$.

Lemma 7.3 (Integral row and column invariance). *If $L \in \text{GL}_m(\mathcal{O})$ and $R \in \text{GL}_n(\mathcal{O})$, then $\Delta_k(LAR) = \Delta_k(A)$ for every k .*

Proof. By the finite Cauchy–Binet formula, every k -minor of LA is a sum of k -minors of A multiplied by k -minors of L . The latter coefficients are integral, so every summand has valuation at least $\Delta_k(A)$. The ultrametric inequality gives $\Delta_k(LA) \geq \Delta_k(A)$. Apply the same argument to L^{-1} for the reverse inequality. Right multiplication is identical, or follows by transposition. The argument also covers vanishing minors. $\square$

Theorem 7.4 (Singular scales are determinantal scales). *For a matrix A of rank r and $1 \leq k \leq r$,*

$$\Delta_k(A) = \gamma_1 + \cdots + \gamma_k, \quad \gamma_k = \Delta_k(A) - \Delta_{k-1}(A). \quad (7.4)$$

In particular the finite slopes of the determinantal profile are nondecreasing. The tuple $(\gamma_1, \dots, \gamma_r)$ is a complete invariant for equivalence under integral invertible row and column operations, with the matrix dimensions fixed.

Proof. In an SVD $A = U\Sigma V^*$ both unitary factors and their inverses are integral by Lemma 7.2. Thus Lemma 7.3 reduces the calculation to Σ . Its nonzero k -minors are products of k distinct nonzero singular values. Their minimum valuation is the sum of the k smallest γ_j , proving (7.4).

Write $\sigma_j = t^{\gamma_j} u_j$, where $u_j \in \mathcal{O}^\times$ is a positive unit. Multiplying diagonal rows by u_j^{-1} shows that A is integrally equivalent to the rectangular diagonal matrix $\text{diag}(t^{\gamma_1}, \dots, t^{\gamma_r}, 0, \dots)$. Thus two matrices with the same tuple are equivalent. Conversely, equivalence preserves Δ_k , hence the tuple and rank. $\square$

For integral matrices this is the singular-value interpretation of their valuation-ring invariant factors. The valuation ring need not be Noetherian or a discrete valuation ring. Only finitely many entries and minors are involved, so a minimum valuation among them exists. Unit equivalence remembers less than unitary equivalence: the latter preserves all singular values, not merely their valuations.

An exterior-power interpretation is also useful. Give $\bigwedge^k K^n$ the inner product for which wedges of coordinate basis vectors are orthonormal. The entries of $\bigwedge^k A$ are the k -minors of A . Wedges of the singular bases yield

$$\left\| \bigwedge^k A \right\|_{\text{op}} = \sigma_1 \cdots \sigma_k.$$

Applying [Lemma 7.1](#) recovers (7.4). Thus a determinantal scale is the scale of the maximal k -dimensional volume factor.

7.3 Why entrywise scales alone are not enough

Example 7.5 (A nearly dependent matrix). Let $q \in \Gamma_{>0}$ and

$$A_q = \begin{pmatrix} 1 & 1 \\ 1 & 1 + t^q \end{pmatrix}.$$

Every entry has valuation zero. Nevertheless, $\det A_q = t^q$, so $\Delta_1 = 0$, $\Delta_2 = q$, and the singular scales are $0, q$. In contrast, the matrix of all ones has the same entrywise valuations and rank one. Changing q changes the hidden singular scale without changing any of those entrywise valuations.

The matrix A_q is real symmetric and positive definite, since its leading principal minors are 1 and t^q . Its singular values are its positive eigenvalues:

$$\sigma_{\pm} = \frac{2 + t^q \pm \sqrt{4 + t^{2q}}}{2}.$$

Hence $\text{lc}(\sigma_+) = 2$ and $\sigma_- = (1/2)t^q + \text{terms of valuation greater than } q$. A minimum-weight assignment using only entry valuations could miss the cancellation in the determinant. Exact minors or an equivalent elimination calculation are needed.

8 Gram coefficients recover scales and leading amplitudes

The polynomial report already develops Newton profiles of scalar polynomials [2]. Positive Gram matrices give an especially clean matrix input to that machinery: their characteristic coefficients carry no leading cancellation from squared minors.

Theorem 8.1 (Positive Cauchy–Binet identity). *For $A \in K^{m \times n}$ write*

$$\det(I_n + sA^*A) = \sum_{k=0}^p c_k s^k, \quad c_0 = 1. \quad (8.1)$$

Then

$$c_k = \sum_{|I|=|J|=k} |\det A_{I,J}|^2 = e_k(\sigma_1^2, \dots, \sigma_r^2). \quad (8.2)$$

For $1 \leq k \leq r$,

$$c_k > 0, \quad v(c_k) = 2\Delta_k(A). \quad (8.3)$$

For $k > r$, $c_k = 0$.

Proof. Expansion of $\det(I + sB)$ by the columns in which an entry of sB is selected shows that its coefficient of s^k is the sum of the principal k -minors of B . For $B = A^*A$, Cauchy–Binet expands the principal minor indexed by J as

$$\det((A^*A)_{J,J}) = \sum_{|I|=k} \overline{\det A_{I,J}} \det A_{I,J}.$$

This proves the first equality in (8.2). Diagonalizing A^*A gives the second. If a k -minor is nonzero, the leading coefficient of the sum of its squared moduli and all other squared minors of minimum valuation is a positive real number. Therefore no leading cancellation is possible, and the valuation is twice the minimum minor valuation. $\square$

Thus the Gram determinant polynomial provides all the singular scales without finding an eigenvector. Its nonzero coefficient valuations are $2\Delta_k$, with successive differences $2\gamma_k$. This statement works in any ordered group Γ ; it does not require drawing a polygon in an ordinary real plane.

8.1 Resolving singular values at a common scale

Define

$$C_k = \text{st}(t^{-2\Delta_k} c_k) > 0 \quad (0 \leq k \leq r), \quad C_0 = 1.$$

Let a maximal equal-scale block be

$$\gamma_{p_0+1} = \cdots = \gamma_{p_0+d} = \gamma,$$

where the preceding scales, if any, are strictly smaller and the following ones, if any, are strictly larger. Write $a_j = \text{lc}(\sigma_j) > 0$.

Theorem 8.2 (Residual polynomial for a singular-scale block). *For $0 \leq \ell \leq d$,*

$$\frac{C_{p_0+\ell}}{C_{p_0}} = e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2). \quad (8.4)$$

Consequently the ordinary real polynomial

$$R_\gamma(X) = X^d + \sum_{\ell=1}^d (-1)^\ell \frac{C_{p_0+\ell}}{C_{p_0}} X^{d-\ell} \quad (8.5)$$

has roots precisely the squared leading coefficients in that block, with multiplicity.

Proof. In the elementary symmetric sum for $c_{p_0+\ell}$, a product of minimum valuation must contain all the p_0 singular values of smaller valuation, exactly ℓ singular values from the block, and none from later blocks. Indeed, omitting a strictly smaller scale while using a larger one increases the sum of valuations; the increasing ordering shows that only choices across the equal boundary can tie the minimum. Therefore its leading coefficient is

$$\left(\prod_{j=1}^{p_0} a_j^2 \right) e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2).$$

For $\ell = 0$, the prefactor is exactly C_{p_0} . Division proves (8.4), and the elementary symmetric factorization proves (8.5). $\square$

This does not give all higher coefficients of the singular values. It gives their valuations and first nonzero coefficients; repeated roots of R_γ may require another scale refinement. It is already stronger than knowing only the Newton slopes.

For [Example 7.5](#),

$$c_1 = 4 + 2t^q + t^{2q}, \quad c_2 = t^{2q},$$

so $C_1 = 4$ and $C_2 = 1$. The two one-element blocks give leading squares 4 and $1/4$, hence leading singular coefficients 2 and $1/2$. For a same-scale example,

$$A = \text{diag}(2t^\gamma, 3t^\gamma), \quad C_1 = 13, \quad C_2 = 36,$$

and $R_\gamma(X) = X^2 - 13X + 36 = (X - 4)(X - 9)$. The valuation alone cannot distinguish the factors 2 and 3, but the positive Gram coefficients do.

9 Rank at a chosen scale

9.1 Exact rank and standard-part rank

Corollary 9.1 (Rank after reduction). *If $A \in \mathcal{O}^{m \times n}$ has nonzero singular scales $\gamma_1, \dots, \gamma_r$, then every $\gamma_j \geq 0$ and*

$$\text{rank}_{\mathbb{C}}(\text{st } A) = \#\{j : \gamma_j = 0\}.$$

Proof. By Lemma 7.1, $v(\sigma_1) = v(A) \geq 0$, so all singular values are integral. Reduce an SVD. Its unitary factors reduce to invertible ordinary unitary matrices, and its diagonal entries reduce to nonzero numbers exactly at valuation zero. $\square$

An invertible matrix can therefore have a singular standard part. Conversely, an invertible standard part certifies invertibility of an integral square matrix. The inverse need not be integral in the first situation, but is integral in the second: the latter is exactly the case in which all its singular scales are zero.

9.2 An intrinsic input filtration

Fix $\theta \in \Gamma$ and define an $\mathcal{O}$ -submodule of the input lattice by

$$M_{\theta}(A) = \{x \in \mathcal{O}^n : Ax \in t^{\theta} \mathcal{O}^m\}. \quad (9.1)$$

Its image $\text{st}(M_{\theta}(A))$ under coordinatewise reduction is a $\mathbb{C}$ -linear subspace of $\mathbb{C}^n$. Also define

$$M_{>\theta}(A) = \{x \in \mathcal{O}^n : v(Ax) > \theta\}.$$

These definitions work even when A or t^{θ} is nonintegral; only the input vectors are required to lie in the integral lattice.

Theorem 9.2 (Filtration dimensions). *For every $\theta \in \Gamma$,*

$$\dim_{\mathbb{C}} \text{st}(M_{\theta}(A)) = n - \#\{j : \gamma_j < \theta\}, \quad (9.2)$$

$$\dim_{\mathbb{C}} \text{st}(M_{>\theta}(A)) = n - \#\{j : \gamma_j \leq \theta\}. \quad (9.3)$$

Only the nonzero singular values enter the counts.

Proof. Use an SVD $A = U\Sigma V^*$. Both unitary factors preserve integral lattices and their reductions are invertible. Write $x = Vy$. For $j \leq r$, condition (9.1) becomes

$$v(y_j) \geq \max(0, \theta - \gamma_j).$$

The kernel coordinates $j > r$ are unrestricted in $\mathcal{O}$. The reduction of y_j must be zero exactly when $\gamma_j < \theta$; all other reduced coordinates can be chosen arbitrarily. This gives (9.2). For the strict condition, reduction of y_j vanishes exactly when $\gamma_j \leq \theta$. Multiplication by the invertible $\text{st } V$ preserves the dimension, proving (9.3). $\square$

The two formulas deliberately differ at the boundary. A direction of singular scale exactly θ remains visible when the output threshold is strict, and disappears if only outputs of strictly larger magnitude are counted. It is helpful to fix explicit notation:

$$r_{<\theta}(A) = \#\{j : \gamma_j < \theta\}, \quad r_{\leq\theta}(A) = \#\{j : \gamma_j \leq \theta\}.$$

These are not replacements for algebraic rank. They describe how much rank is visible at a specified valuation resolution. In the integral case, $r_{\leq 0}(A) = \text{rank}(\text{st } A)$.

Example 9.3 (Three genuinely different levels). Choose a workspace containing 1 and ω in its value group and let

$$A = \text{diag}(1, t^2, t^\omega), \quad t = \omega^{-1}.$$

Its exact rank is three and its standard-part rank is one. Its closed-scale rank $r_{\leq \theta}$ is one for $0 \leq \theta < 2$, two for $2 \leq \theta < \omega$, and three for $\theta \geq \omega$. No finite integer cutoff N reaches the last direction. This is not a statement about a slow numerical convergence rate; it is an ordered-group fact, $N < \omega$ for every ordinary N .

10 A root-free scale algorithm and its computational contract

10.1 Minimum-valuation elimination

The determinantal formula is finite, but enumerating all minors is expensive. A valuation-aware elimination computes the same scale tuple without singular vectors or algebraic root extraction.

Proposition 10.1 (Exact scale elimination). *Suppose exact field arithmetic, exact zero testing, and comparison of valuations of nonzero elements are available. The following algorithm terminates after at most $\min(m, n)$ pivots and returns $\gamma_1, \dots, \gamma_r$ in nondecreasing order.*

In the active submatrix, choose a nonzero entry of minimum valuation and move it to the leading position by row and column swaps. Clear the remaining entries in its column by row operations, and then clear its row by column operations. Record its valuation and recurse on the trailing submatrix. Stop when the active submatrix is zero.

Proof. Write one active block as

$$\begin{pmatrix} a & b \\ c & D \end{pmatrix}, \quad v(a) = \alpha \leq v(\text{every entry}).$$

The row multipliers c_i/a and column multipliers b_j/a are integral. Their elementary matrices and inverses therefore lie in $\text{GL}(\mathcal{O})$. The two clearings leave the Schur complement

$$D - ca^{-1}b. \tag{10.1}$$

Every term in it has valuation at least α , because $v(c_i) + v(b_j) - v(a) \geq \alpha$. Cancellation can raise the valuation, but cannot lower it. Thus subsequent pivot valuations are nondecreasing. After finitely many steps the result is a diagonal matrix and a zero block. All transformations were integral and invertible, so they preserve the Δ_k . The diagonal pivot valuations must therefore be their successive differences, which are the singular scales by [Theorem 7.4](#). $\square$

The algorithm uses $O(mn \min(m, n))$ field operations, apart from pivot comparisons. This counts algebraic operations; it is not a bit-complexity bound for arbitrary Hahn-series representations. A single exact zero test may already be unavailable for a given symbolic representation. Moreover, the elimination matrices are generally not unitary: this algorithm returns the singular *scales*, not the singular values or the best low-rank approximation itself.

10.2 What must be provided by a symbolic scalar representation

The algorithm has four logically separate requirements: a representation closed under the field operations it performs, exact zero testing, exact leading-exponent access for nonzero outputs, and an effective order on those exponents. Rational functions over a finite rational or Gaussian-rational monomial field can meet these requirements. A name for an arbitrary surreal number

does not automatically meet any of them. This is the same kind of denotation-versus-computation distinction emphasized by the repository's computer-algebra strand [1].

Cancellation is the principal practical issue. A representation that stores only the current leading term is insufficient: (10.1) may cancel it. Correct implementations must either compute further exact coefficients, attach a certified remainder bound, or report that the pivot scale is unresolved. Reporting a zero pivot merely because all currently stored coefficients vanish is unsound.

10.3 Why finite truncation cannot certify every exact rank

Fix any precision threshold θ . Choose $\eta > \theta$. The matrices

$$\text{diag}(1, 0), \quad \text{diag}(1, t^\eta)$$

agree at all exponents at most θ but have different exact ranks. Thus finite-precision coefficient access alone cannot decide exact rank uniformly. An exact representation may still decide it, for example by computing a nonzero symbolic determinant, but truncation alone cannot.

The right numerical-style output is therefore often a certified rank profile up to a declared cutoff, with the remaining directions unresolved. The perturbation theorem in Section 11 specifies exactly what precision is sufficient for such a certificate. A finite exact rational kernel for testing the elimination and Gram identities is included in the archive; its scope is described in Section B.

11 Condition numbers and perturbation certificates

11.1 Distance to singularity is a minimum, not merely an infimum

This subsection again holds over any real closed F and $K = F(i)$.

Theorem 11.1 (Distance to singularity). *For an invertible square A ,*

$$\|A^{-1}\|_{\text{op}} = \sigma_n(A)^{-1}, \quad \min_{A+E \text{ singular}} \|E\|_{\text{op}} = \sigma_n(A). \quad (11.1)$$

The latter minimum is attained by $E = -\sigma_n u_n v_n^$ for any matched least-singular vector pair.*

Proof. Invert the SVD to obtain the first formula. The stated perturbation deletes one diagonal entry of Σ and has norm σ_n . Conversely, if $A + E$ is singular, choose a unit vector x in its kernel. Then

$$\|E\|_{\text{op}} \geq \|Ex\| = \|Ax\| \geq \sigma_n.$$

□

Define the ordered spectral condition number by

$$\kappa(A) = \|A\|_{\text{op}} \|A^{-1}\|_{\text{op}} = \frac{\sigma_1}{\sigma_n} \geq 1.$$

It is an element of F , not necessarily an ordinary real. In a Hahn workspace,

$$v(\kappa(A)) = \gamma_1 - \gamma_n. \quad (11.2)$$

Consequently $\kappa(A)$ is infinite precisely when $\gamma_1 < \gamma_n$. Matrices whose nonzero singular values all have the same valuation have finite condition number, although its ordinary leading coefficient can still be large. Overall scale and relative conditioning are separate: $t^{-\omega}I$ has infinite norm but condition number one.

Theorem 11.2 (Inverse and linear-system perturbation). *Let A be invertible and put $q = \|A^{-1}\|_{\text{op}} \|E\|_{\text{op}}$. If $q < 1$, then $A + E$ is invertible,*

$$\|(A + E)^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}}{1 - q}, \quad (11.3)$$

$$\|(A + E)^{-1} - A^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}^2 \|E\|_{\text{op}}}{1 - q}. \quad (11.4)$$

If $Ax = b \neq 0$ and $(A + E)(x + \delta x) = b + f$, then

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - q} \left(\frac{\|f\|}{\|b\|} + \frac{\|E\|_{\text{op}}}{\|A\|_{\text{op}}} \right). \quad (11.5)$$

Proof. For every vector x ,

$$\|(A + E)x\| \geq (\sigma_n(A) - \|E\|_{\text{op}}) \|x\|.$$

The coefficient is positive under the hypothesis. The map is injective, hence invertible in finite dimension, and this inequality gives (11.3). The finite resolvent identity

$$(A + E)^{-1} - A^{-1} = -(A + E)^{-1}EA^{-1}$$

gives (11.4). Also $\delta x = (A + E)^{-1}(f - Ex)$. Apply (11.3), divide by $\|x\|$, and use $\|b\| \leq \|A\|_{\text{op}} \|x\|$ to obtain (11.5). $\square$

No geometric series was used. This matters because the ordered inequality $q < 1$ is not by itself a Hahn-summability hypothesis.

11.2 A certified precision theorem for singular scales

Return to a Hahn workspace. When $v(A) = \eta$, every entry of A has valuation at least η ; this is the matrix precision convention below.

Theorem 11.3 (Preserving visible singular data). *Let $A, B = A + E$ have the same dimensions. If $v(E) > \gamma_k(A)$ for a nonzero singular value $\sigma_k(A)$, then for all $1 \leq j \leq k$,*

$$v(\sigma_j(B)) = \gamma_j(A), \quad \text{lc}(\sigma_j(B)) = \text{lc}(\sigma_j(A)).$$

More generally, if $v(E) > \theta$, then

$$r_{\leq \theta}(A) = r_{\leq \theta}(B), \quad r_{< \theta}(A) = r_{< \theta}(B).$$

The exact ranks need not be equal.

Proof. By Lemma 7.1, $v(\|E\|_{\text{op}}) = v(E)$ for $E \neq 0$. The ordered singular-value estimate (5.8) therefore implies

$$v(\sigma_j(B) - \sigma_j(A)) \geq v(E).$$

An error of strictly larger valuation cannot alter a nonzero leading term. This proves the first assertion. If a singular value has valuation greater than θ , or is zero, adding an error of valuation greater than θ keeps it above that threshold. Singular values at or below the threshold retain their leading terms. Apply this coordinatewise to the ordered singular lists for both asserted counts. The case $E = 0$ is immediate. $\square$

Strictness is essential. For $A = \text{diag}(1, t^\theta)$ with $\theta > 0$, a perturbation $E = \text{diag}(0, -t^\theta)$ has exactly valuation θ and destroys the second singular direction. Even under the strict hypothesis, a zero singular value can become a nonzero value of a still smaller magnitude; $\text{diag}(1, 0)$ and $\text{diag}(1, t^\eta)$ with $\eta > \theta$ show why exact rank is not certified. The theorem certifies the rank *visible at the stated scale*.

12 Nonnormal matrices and ordered pseudospectra

Eigenvalues alone do not control a general matrix. For example,

$$N_L = \begin{pmatrix} 0 & L \\ 0 & 0 \end{pmatrix}$$

has only the eigenvalue zero, while $\|N_L\|_{\text{op}} = |L|$ can be infinite. The distinction between eigenvalues and singular values is as important in the surcomplex field as it is over $\mathbb{C}$.

For a square matrix and an ordered tolerance $\varepsilon \in F_{\geq 0}$, define the closed ordered pseudospectrum by

$$\Lambda_\varepsilon(A) = \{z \in K : \sigma_n(zI - A) \leq \varepsilon\}. \quad (12.1)$$

Proposition 12.1 (Exact perturbation interpretation). *A scalar z belongs to $\Lambda_\varepsilon(A)$ if and only if there is E with $\|E\|_{\text{op}} \leq \varepsilon$ for which z is an eigenvalue of $A + E$. If $z \notin \text{spec } A$ and $\varepsilon > 0$, this is equivalent to*

$$\|(zI - A)^{-1}\|_{\text{op}} \geq \varepsilon^{-1}.$$

For normal A ,

$$\Lambda_\varepsilon(A) = \bigcup_{\lambda \in \text{spec } A} \{z : |z - \lambda| \leq \varepsilon\}. \quad (12.2)$$

Proof. Apply the distance-to-singularity argument to $zI - A$. If it is already singular, use $E = 0$. Otherwise [Theorem 11.1](#) gives an attained perturbation of norm $\sigma_n(zI - A)$ making it singular; changing its sign converts it to a perturbation of A . The lower bound follows by applying $zI - A$ to an eigenvector of $A + E$. The inverse characterization is [\(11.1\)](#). For normal A , unitary diagonalization gives singular values of $zI - A$ equal to the numbers $|z - \lambda_j|$, proving [\(12.2\)](#). $\square$

Example 12.2 (An infinitesimal perturbation with a larger spectral effect). Let

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ t^2 & 0 \end{pmatrix}.$$

Then $\|E\|_{\text{op}} = t^2$, whereas $N + E$ has eigenvalues $\pm t$. Thus $t \in \Lambda_{t^2}(N)$ although its distance from the unperturbed spectrum is $t > t^2$. More generally, for nonzero infinitesimal z , $zI - N$ has $\Delta_1 = 0$ and $\Delta_2 = 2v(z)$, so its least singular value has valuation $2v(z)$. The nonlinear sensitivity is visible directly in the determinantal scale profile of the resolvent matrix.

Here ε is an actual ordered magnitude, not just a value-group element. A valuation threshold gives a coarser notion. In particular, replacing the ordered disks in [\(12.2\)](#) by valuation balls would change the assertion. No fine-continuous contour is needed in the pseudospectral argument.

13 Spectral subspaces and infinitesimal gaps

Individual eigenvectors are not canonical, especially at multiplicities. Orthogonal spectral projectors are the better perturbation objects. The following finite-dimensional Frobenius version of a Davis–Kahan bound is proved directly over F ; the classical context is [\[10, 11\]](#). No inverse trigonometric function is needed to measure the subspace displacement.

Theorem 13.1 (A separated-subspace bound). *Let $H = H^*$, $B = H + E = B^*$, and let P be an orthogonal projector onto an H -invariant subspace. Let Q project onto a B -invariant subspace of rank k . Suppose that every eigenvalue λ of $H|_{\text{im}(I-P)}$ and every eigenvalue μ of $B|_{\text{im } Q}$ satisfy*

$$|\lambda - \mu| \geq \delta > 0.$$

Then

$$\|(I - P)Q\|_F \leq \frac{\|E\|_F}{\delta}. \quad (13.1)$$

If $\text{rank } P = \text{rank } Q$, also

$$\|P - Q\|_F \leq \frac{\sqrt{2}\|E\|_F}{\delta}. \quad (13.2)$$

Proof. Choose orthonormal eigenbasis matrices Y for $\text{im}(I - P)$ and X for $\text{im } Q$, so $HY = Y\Lambda$ and $BX = XM$, with real diagonal Λ, M . Put $C = Y^*X$. Multiplying $HX + EX = XM$ on the left by Y^* gives the Sylvester equation

$$\Lambda C - CM = -Y^*EX.$$

Entrywise, $(\lambda_a - \mu_b)c_{ab} = -(Y^*EX)_{ab}$, hence

$$\|C\|_F^2 \leq \delta^{-2} \|Y^*EX\|_F^2 \leq \delta^{-2} \|E\|_F^2.$$

The last inequality follows because X and Y have orthonormal columns; extend them to unitary matrices and observe that taking a submatrix cannot increase the sum of squared moduli. Since $(I - P)Q = YCX^*$, $\|(I - P)Q\|_F = \|C\|_F$. This proves (13.1).

For equal ranks,

$$\|P - Q\|_F^2 = \text{tr } P + \text{tr } Q - 2\text{tr}(PQ) = 2\|(I - P)Q\|_F^2,$$

which gives (13.2). $\square$

Corollary 13.2 (Using an unperturbed cluster gap). *Select a nonempty proper set of eigenvalue indices of H , separated from all complementary indices by at least $g > 0$. Let P project onto this cluster, and let Q select the same indices in the decreasing eigenvalue list of $H + E$. If $e = \|E\|_{\text{op}} < g/2$, then*

$$\|P - Q\|_F \leq \frac{\sqrt{2}\|E\|_F}{g - e}.$$

Proof. The eigenvalue estimate in Theorem 5.2 moves each selected eigenvalue by at most e . Its distance from an unselected eigenvalue of H is at least $g - e$. No eigenvalue can cross the external gap under $e < g/2$. The chosen subspaces have the same dimension, so apply Theorem 13.1 with $\delta = g - e$. $\square$

The Frobenius separation condition is all that was used in the proof. We are not asserting every operator-norm variant of the classical Davis–Kahan theorem under an arbitrary separation hypothesis. The Frobenius estimate is sufficient for an important surreal consequence: if $v(E) > v(g)$ in a Hahn workspace, then the projector displacement is infinitesimal. Indeed, $\|E\|_F/g$ then has positive valuation, so $e < g/2$ and the displayed bound has positive valuation.

Example 13.3 (Infinitesimal error, ordinary subspace rotation). Consider

$$H = \begin{pmatrix} 0 & 0 \\ 0 & t^2 \end{pmatrix}, \quad B = \frac{t^2}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

They have the same eigenvalues $0, t^2$. Their zero-eigenspace projectors are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Exact multiplication gives

$$\|B - H\|_{\text{op}} = \frac{t^2}{\sqrt{2}}, \quad \|P - Q\|_{\text{F}} = 1, \quad \|P - Q\|_{\text{op}} = \frac{1}{\sqrt{2}}.$$

Thus the matrix error is infinitesimal but the subspace change is not. Its size is comparable to the gap t^2 , so the gap-relative smallness hypothesis fails. Discarding the common infinitesimal factor before asking about eigenvectors would conceal the entire phenomenon.

13.1 Singular subspaces and a warning about Gram squaring

Right singular subspaces are eigenspaces of A^*A . Under $A \mapsto A + E$,

$$(A + E)^*(A + E) - A^*A = A^*E + E^*A + E^*E,$$

so its operator norm is at most $2\|A\|_{\text{op}}\|E\|_{\text{op}} + \|E\|_{\text{op}}^2$. One can therefore apply [Theorem 13.1](#) to squared-singular-value gaps. This is valid but can be quantitatively wasteful: it changes gaps $\sigma_i - \sigma_j$ into $\sigma_i^2 - \sigma_j^2$. A direct singular-subspace or Hermitian-dilation argument can yield better bounds, but no sharp Wedin theorem is claimed here. The distinction is particularly relevant when both singular values are infinitesimal.

14 Degenerate clusters and Schur-complement corrections

A cluster of repeated or nearly repeated eigenvalues cannot generally be understood by retaining only the perturbation inside that cluster. Coupling to its complement can create a smaller, second-order splitting. The right starting point is an exact finite identity.

Proposition 14.1 (Exact effective matrix). *Suppose a Hermitian matrix is written in an orthogonal decomposition as*

$$H = \begin{pmatrix} \lambda_0 I + E_{11} & E_{12} \\ E_{21} & D + E_{22} \end{pmatrix}, \quad E_{21} = E_{12}^*.$$

For $z \in K$ such that $D + E_{22} - zI$ is invertible,

$$\det(H - zI) = \det(D + E_{22} - zI) \det S(z), \tag{14.1}$$

$$S(z) = (\lambda_0 - z)I + E_{11} - E_{12}(D + E_{22} - zI)^{-1}E_{21}. \tag{14.2}$$

If the ordered distance from z to every eigenvalue of $D + E_{22}$ is at least $\delta > 0$, then

$$\|E_{12}(D + E_{22} - zI)^{-1}E_{21}\|_{\text{op}} \leq \frac{\|E_{12}\|_{\text{op}}^2}{\delta}. \tag{14.3}$$

Proof. Multiply $H - zI$ on the left by the determinant-one block matrix

$$\begin{pmatrix} I & -E_{12}(D + E_{22} - zI)^{-1} \\ 0 & I \end{pmatrix}.$$

The resulting matrix is block lower triangular with diagonal blocks $S(z)$ and $D + E_{22} - zI$, proving (14.1). Diagonalization of the Hermitian complement shows that its shifted inverse has norm at most δ^{-1} . Apply the product norm bounds and $\|E_{21}\|_{\text{op}} = \|E_{12}\|_{\text{op}}$. $\square$

For real z outside the complementary spectrum, $S(z)$ is Hermitian. It is an eigenvalue-dependent effective matrix, not in general one fixed matrix whose eigenvalues already solve the original problem. Freezing z or dropping the inverse correction requires a separate error estimate. Formula (14.3) supplies one such estimate: the correction has the scale of the square of the coupling divided by the complementary gap. If that gap is itself infinitesimal, a seemingly second-order term can be as important as a first-order term.

Example 14.2 (A spectral shift invisible in the small diagonal block). Let

$$H_t = \begin{pmatrix} 0 & t \\ t & 1 \end{pmatrix}.$$

Its eigenvalues and low-eigenvalue projector are exactly

$$\lambda_{\pm} = \frac{1 \pm \sqrt{1 + 4t^2}}{2}, \quad (14.4)$$

$$P_- = \frac{\lambda_+ I - H_t}{\lambda_+ - \lambda_-}. \quad (14.5)$$

The small eigenvalue is

$$\lambda_- = -t^2 + t^4 - 2t^6 + 5t^8 + O_v(t^{10}). \quad (14.6)$$

Here $O_v(t^{10})$ denotes a Hahn remainder of valuation at least 10; t is not a real variable tending to zero. The expansion follows either from the binomial series justified in Section 15, or by substituting the displayed finite polynomial into $\lambda^2 - \lambda - t^2$ and checking the remainder order, then selecting the root with positive valuation.

The direct perturbation inside the first coordinate is zero. Nevertheless, (14.2) gives the scalar equation $-z - t^2/(1 - z) = 0$, whose leading solution is $z = -t^2$. An eigenvector may be taken as $(1, \lambda_-/t)^T$, displaying an order- t mixing with a spectral shift of order t^2 .

The companion matrix

$$\tilde{H}_t = \begin{pmatrix} t^2 & t \\ t & 1 \end{pmatrix} = \begin{pmatrix} t \\ 1 \end{pmatrix} \begin{pmatrix} t & 1 \end{pmatrix}$$

is positive semidefinite of exact rank one. In its effective scalar at $z = 0$, the direct t^2 term and the coupling correction t^2 cancel exactly. The difference between a small eigenvalue and a zero eigenvalue can therefore depend on cancellation between terms of the same scale. Neither leading entry valuations nor a generic “second-order is negligible” rule resolves it.

15 Finite functional calculus versus infinite Hahn series

15.1 Finite spectral calculus needs no convergence

Let H be Hermitian with distinct eigenvalues in a finite set S . For any prescribed scalars $f(\lambda) \in K$, define

$$f(H) = \sum_{\lambda \in S} f(\lambda) P_{\lambda}. \quad (15.1)$$

Lagrange interpolation gives a polynomial $p \in K[X]$ of degree less than $|S|$ with $p(\lambda) = f(\lambda)$; thus $f(H) = p(H)$. The construction preserves finite sums and products and depends only on the values of f at the spectrum. For real-valued spectral data it gives a Hermitian matrix; for nonnegative data it gives a positive semidefinite matrix. It also yields

$$\|f(H)\|_{\text{op}} = \max_{\lambda \in S} |f(\lambda)|.$$

For $z \notin S$, the resolvent is

$$(zI - H)^{-1} = \sum_{\lambda \in S} \frac{P_\lambda}{z - \lambda}.$$

All of this is finite algebra. It does not assert that an arbitrary scalar function already has a chosen surreal extension. In particular, one cannot use (15.1) to define unspecified values such as a global surreal sine at an infinite argument by circular reasoning. One must first supply the values to be interpolated.

15.2 A genuine matrix Hahn-summability theorem

A family of Hahn series is *strongly summable* if the union of its supports is well ordered and every exponent occurs in only finitely many members. Its sum is then defined coefficientwise by finite sums. For matrices this means entrywise strong summability, equivalently the same condition on the finitely many entry families together.

We use the standard support lemma underlying Hahn-field arithmetic: if $S \subseteq \Gamma_{>0}$ is well ordered, its additive monoid

$$S^* = \{0\} \cup \{s_1 + \cdots + s_k : k \geq 1, s_j \in S\}$$

is well ordered, and every exponent has only finitely many representations as an ordered finite sum of members of S . This is the positive-support form of the Neumann lemma; see the support development in Gonshor, Chapter 5A [4]. The lemma does not assert that $k\rho$ is cofinal in Γ for a positive ρ .

Theorem 15.1 (Matrix inverse and square-root series). *If $E \in K_\Gamma^{n \times n}$ has $v(E) > 0$, then*

$$\sum_{k \geq 0} E^k \tag{15.2}$$

is strongly Hahn-summable and equals $(I - E)^{-1}$. If in addition $E = E^$, then*

$$\sum_{k \geq 0} \binom{1/2}{k} E^k \tag{15.3}$$

is strongly Hahn-summable and equals the positive square root of $I + E$.

Proof. The finite union of the entry supports of E is a well-ordered subset $S \subseteq \Gamma_{>0}$. An entry of E^k is a finite sum of products of k entries, so its support is contained in sums of k elements of S . The support lemma gives a well-ordered union and finitely many contributing lengths and tuples at every exponent. For a fixed length there are only finitely many matrix-index paths. This proves strong summability of both series, and justifies the coefficientwise rearrangements below.

The finite formal identity $(1 - X) \sum_{k \geq 0} X^k = 1$ is valid coefficientwise after substituting E , giving a two-sided inverse. Likewise the formal binomial identity makes the square of (15.3) equal to $I + E$. Its sum R is Hermitian, because every power of E is, and $v(R - I) > 0$. The norm of $R - I$ is therefore infinitesimal by Lemma 7.1; all eigenvalues of R are positive by Theorem 5.2. Uniqueness of the positive square root from Proposition 4.3 identifies R with $(I + E)^{1/2}$. $\square$

15.3 Two failures of a familiar shortcut

The ordered condition $\|E\|_{\text{op}} < 1$ does not imply strong Hahn-summability of (15.2). For $E = \frac{1}{2}I$, every nonzero entry of every power has exponent zero, so infinitely many members meet the same exponent. The Hahn summability criterion fails, although $(I - E)^{-1} = 2I$ exists algebraically. Ordinary analytic summation of real coefficients is a different operation.

Even when the Hahn series exists, it need not be a topological limit of its partial sums in the full surreal field. With $E = tI$ and $t = \omega^{-1}$,

$$(I - tI)^{-1} - \sum_{k=0}^N t^k I = \frac{t^{N+1}}{1 - t} I.$$

For every ordinary N , this error is larger in modulus than t^ω . Thus the partial sums fail the fine-surreal convergence test at the single tolerance t^ω .

There is no contradiction with internal valuation convergence in $K_{\mathbb{Q}} = \mathbb{C}((t^{\mathbb{Q}}))$: the natural-number exponents are cofinal in $\mathbb{Q}$, so these same partial sums do converge in that field's own valuation topology. In the larger workspace with value group $\mathbb{Q} + \mathbb{Q}\omega$, they do not, because the threshold ω is now present. Algebraic identities, strong summability, and workspace-relative topology are three different claims; only the first two were needed for [Theorem 15.1](#).

16 Two applications: regularization and finite tensors

16.1 Regularization as an explicit scale filter

For $\alpha > 0$, consider the regularized least-squares objective

$$J_\alpha(x) = \|Ax - b\|^2 + \alpha^2 \|x\|^2.$$

The matrix $A^*A + \alpha^2 I$ is positive definite, regardless of the rank of A . Define

$$x_\alpha = (A^*A + \alpha^2 I)^{-1} A^*b. \quad (16.1)$$

Expanding and using its normal equation yields

$$J_\alpha(x) = J_\alpha(x_\alpha) + (x - x_\alpha)^*(A^*A + \alpha^2 I)(x - x_\alpha).$$

Hence (16.1) is the unique minimizer over the whole field, with no compactness or coercivity argument required.

In singular coordinates,

$$x_\alpha = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \alpha^2} (u_j^* b) v_j, \quad Ax_\alpha = \sum_{j=1}^r \frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} (u_j^* b) u_j. \quad (16.2)$$

The fitted-output factor has a particularly clear reduction. Put $\beta = v(\alpha)$ and $\gamma = v(\sigma_j)$, with positive leading coefficients $b_0 = \text{lc}(\alpha)$ and $a_0 = \text{lc}(\sigma_j)$. Positivity of the denominator gives

$$\text{st} \left(\frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} \right) = \begin{cases} 1, & \gamma < \beta, \\ \frac{a_0^2}{a_0^2 + b_0^2}, & \gamma = \beta, \\ 0, & \gamma > \beta. \end{cases} \quad (16.3)$$

The quotient always lies in $[0, 1]$, so its standard part exists. This is a scale filter: larger singular directions pass, smaller ones disappear in the ordinary shadow, and the boundary scale retains a nontrivial leading amplitude. At that boundary, a valuation-only description is insufficient.

Proposition 16.1 (Ordinary data and infinitesimal regularization). *If A and b have ordinary complex entries and α is a positive infinitesimal in a Hahn workspace, then x_α is finite and*

$$\text{st}(x_\alpha) = A^\dagger b,$$

where the right side is the ordinary complex pseudoinverse solution.

Proof. Choose an ordinary complex SVD. Every nonzero singular value is a positive ordinary real number. In (16.2), the coefficient $\sigma_j/(\sigma_j^2 + \alpha^2)$ is finite and has standard part $1/\sigma_j$. There are finitely many terms, so reduction gives the pseudoinverse formula. $\square$

The ordinary-data hypothesis matters. For $A = (t)$, $b = 1$, and $\alpha = t^2$, the regularized solution is $t^{-1}/(1 + t^2)$, which is infinite and has no standard part. Nor does a universal choice of one infinitesimal α resolve every surreal scale: Example 9.3 already exhibits a direction smaller than all finite powers of a chosen basic infinitesimal.

16.2 Schmidt decomposition and infinitesimal components

Give the finite tensor space $K^m \otimes_K K^n$ the inner product for which the coordinate tensor basis is orthonormal. A vector

$$\psi = \sum_{a,b} A_{ab} e_a \otimes f_b$$

is identified with its coefficient matrix A . The SVD immediately gives

$$\psi = \sum_{j=1}^r \sigma_j u_j \otimes \bar{v}_j, \quad \|\psi\|^2 = \sum_{j=1}^r \sigma_j^2. \quad (16.4)$$

Conjugation on the second vector is required by the convention $A = U\Sigma V^*$. The least number of simple tensors in a representation of ψ is its matrix rank: a sum of k simple tensors gives a matrix of rank at most k , and (16.4) attains $k = r$.

For a unit vector, the eigenvalues of the positive trace-one matrix AA^* are the σ_j^2 and zeros. Thus the finite algebraic parts of Schmidt rank and reduced-matrix spectral calculations make sense over K . For example,

$$\psi_t = \frac{e_0 \otimes f_0 + t e_1 \otimes f_1}{\sqrt{1 + t^2}}$$

has exact tensor rank two, whereas its standard part is the rank-one tensor $e_0 \otimes f_0$. Its second squared Schmidt coefficient is the positive infinitesimal $t^2/(1 + t^2)$, not zero. This illustrates precisely what is lost on taking an ordinary shadow.

This is a finite algebraic application, not a proposed physical interpretation of surreal probabilities. No infinite tensor product, measure-valued probability theory, entropy extension, or quantum dynamics is supplied by these calculations.

17 What the chapter establishes, and what remains separate

The finite-dimensional order theory is robust: orthogonality, spectral and singular-value decompositions, least squares, low-rank optimization, and the perturbation bounds proved above require real closedness, not Archimedean completeness. Their proofs exhibit the relevant witnesses, so questions about class-sized compactness never arise.

The scale theory contains additional information not visible in an ordinary complex matrix. Nonzero singular values have a finite ordered list of valuations. Determinantal minima recover that list, positive Gram coefficients recover its leading amplitudes in blocks, and the input filtration gives an intrinsic meaning to rank at a chosen resolution. The same data govern conditioning and perturbation certificates. The minimum-valuation elimination gives a finite algebraic route to those invariants when the scalar representation supplies the required exact operations.

Several nearby subjects remain distinct. There is no theorem here about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$, bounded-operator spectra, compact or trace-class operators, or spectral

measures. There is no global analytic choice of eigenvectors through collisions or across multiple parameters. The pointwise spectral theorem alone does not produce a support-controlled Hahn expansion for an entire eigenbasis depending on parameters. Nor do the exact decompositions automatically provide effective algorithms for arbitrary surreal inputs. Finally, the strong matrix series of [Theorem 15.1](#) is not a claim of fine-topological convergence.

For the repository, a natural placement would be `docs/surcomplex/spectral-theory/`. The chapter should follow the workspace conventions of the foundations and polynomial-algebra reports, and can be read independently of the analytic coefficient-ring machinery. The most useful cross-reference from polynomial algebra is the bridge from its Newton profiles to [Theorems 8.1](#) and [8.2](#); the most useful cross-reference from computer algebra is the exact-operation contract in [Section 10](#). No repository files were modified in preparing this article.

A A proof and hypothesis ledger

Result	Scalar hypotheses	Essential finite mechanism
Spectral theorem and SVD	Real closed F , $K = F(i)$	Algebraic roots, positivity, finite orthogonal complements
Variational and low-rank extrema	Same	Eigenbasis witnesses and dimension intersection, not compactness
Least squares and pseudoinverse	Same	Orthogonal coordinates and finite sums of nonnegative squares
Singular scales from minors	Divisible Hahn workspace	Integral unitary factors and Cauchy–Binet
Leading singular amplitudes	Same, real/complex residue field	Positive sum of squared minor coefficients
Rank filtration	Same, chosen integral lattice	Lattice preservation by unitaries and coordinate reduction
Scale elimination	Exact field and valuation operations	Minimum pivot, integral operations, Schur complement
Inverse perturbation	Real closed F ; $q < 1$	A positive lower singular bound and a finite inverse identity
Visible-rank stability	Hahn workspace; strict precision threshold	Ordered singular-value Lipschitz bound plus leading-term preservation
Spectral-subspace bound	Real closed F ; separated selected/complementary spectra	Entrywise Sylvester equation and Frobenius norm
Infinite matrix series	Hahn workspace; $v(E) > 0$	Positive support lemma, not cofinality of integer multiples

The order-theoretic arguments also apply directly to real symmetric matrices over F . They do not apply to an arbitrary algebraically closed field equipped with an arbitrary involution: positivity of x^*x is the indispensable additional structure. Conversely, the valuation-ring elimination argument itself requires no conjugation or positivity; those structures are what identify its invariant factors with singular values. This separation helps avoid attributing a positivity theorem to general valued-field linear algebra.

B Reproducible finite checks

The archive contains `code/verify_examples.py`, an exact SymPy verification program, together with its recorded JSON and text reports. The valuation and elimination kernel uses rational functions over $\mathbb{Q}(i)$ in two formal positive monomials t, u , with the lexicographic valuation convention

$$v(t) = (0, 1), \quad v(u) = (1, 0).$$

These correspond to the exponents 1 and ω in the subgroup $\mathbb{Z} + \mathbb{Z}\omega$. A rational function is first canceled exactly; the valuation is the least lexicographic monomial exponent of its numerator minus that of its denominator. This finite rational-function field embeds in a divisible Hahn workspace, but the code is not a general Hahn-series engine. Separate finite identity checks also use exact square-root expressions for the two-by-two spectral examples.

The checks compare minimum-valuation elimination with exhaustive minors on small exact matrices, including matrices obtained by integral invertible row and column operations from prescribed diagonal scales. They also check the positive Gram coefficient identity, the residual polynomial examples, Penrose identities for a rank-deficient matrix, the exact projector and rotation examples, the second-order spectral expansion, the nilpotent pseudospectral example, and the regularization and tensor identities. All symbolic assertions must pass for the program to write a successful report; no floating-point sampling is used.

The bundled run passed 176 exact assertions across 18 matrix-profile cases (Python 3.13.5, SymPy 1.14.0). The recorded run is a reproducibility check of these finite identities and of the implementation on its stated test cases. It is not verification of the general mathematical proofs, of arbitrary ordered groups or Hahn supports, or of proper-class foundations. In particular, a finite test cannot prove the support lemma or certify a general surreal equality algorithm. The exact assertion count and software versions are recorded in `data/verification_report.json` rather than being used as a proxy for mathematical correctness.

C Notation at a glance

Symbol	Meaning
$F, K = F(i)$	A real closed ordered field and its algebraic complexification
F_Γ, K_Γ	Real and complex Hahn workspaces, $t^\gamma = \omega^{-\gamma}$
A^*	Conjugate transpose
$\sigma_1 \geq \dots \geq \sigma_r > 0$	Nonzero singular values, in decreasing ordered magnitude
$\ A\ _{\text{op}}, \ A\ _F$	Operator and Frobenius norms, valued in $F_{\geq 0}$
$v(a), \text{lc}(a)$	Least Hahn exponent and coefficient at that exponent
$v(A)$	Minimum valuation of the entries; $v(0) = \infty$
γ_j	$v(\sigma_j)$, in nondecreasing order
Δ_k	Minimum valuation of a k -minor, with $\Delta_0 = 0$
$\mathcal{O}, \mathfrak{m}, \text{st}$	Integral valuation ring, positive-valuation ideal, residue map
c_k	Coefficient of s^k in $\det(I + sA^*A)$
C_k	Positive leading coefficient of c_k , after monomial normalization
$r_{<\theta}, r_{\leq\theta}$	Numbers of nonzero singular scales below or at a cutoff
$A^\dagger$	Moore–Penrose inverse
$\kappa(A)$	Ordered spectral condition number of an invertible matrix
$O_v(t^\eta)$	A formal remainder with valuation at least η , not a real limit

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